

Part B “Project description”

Project title: **National human capital management at the micro-level: research and modelling of processes and tools**

1. Scientific excellence

Human capital is generally understood as the knowledge and skills an individual accumulates over a lifetime, together with their health, all of which enable them to realize their potential as a productive member of society. The quantity and quality of a country’s total human capital – national human capital (hereinafter also HC) – are characterized by the number of employed economically active people and by the levels of their health, education, and skills. Each country primarily focuses on overcoming its own HC needs in line with the requirements and projections of its national labour market; therefore, there is no single internationally accepted definition of HC, including at the EU-27 level. In the 20th century, HC gradually became a key driver of economic growth in every country. A country’s gross domestic product (GDP) can be expressed as follows:

$$GDP = \text{number of employers} * \text{employer's annual productivity}$$

National human capital is formed, developed, and sustained, both quantitatively and qualitatively, through the implementation of a range of national level policies. The quantity of HC is determined primarily by the outcomes of demographic and employment policies. Education policy is a key determinant of its quality, while health policy has a significant impact on both the quantity and quality of HC. Social cohesion policy should aim to consolidate HC and strengthen society’s capacity for collective action, which is a prerequisite for the full use of HC capacity.

In 2025 European Commission identified HC mismatch to the needed growth of European competitiveness on global level, outlining demographic pressures (i.e., number of employers) and slowing productivity in adjacent fields as strategic vulnerabilities, which are serious barriers for urgent unlocking Europe’s growth potential (COM(2025) 959). Under Eurostat’s baseline labour market projection scenario, the economically active population aged 15–74 is expected to decline in 13 EU-27 Member States by 2035 and in 18 Member States by 2050. M. Draghi’s report *The Future of European Competitiveness* highlights the widening productivity gap with the US, describing it as an “existential challenge”.

State of Health in the EU, Synthesis Report 2025 highlights major challenges facing health system – in 2023, more than 950,000 deaths could have been prevented in the EU27, while 19 million potential productive life years were lost (Eurostat), equivalent to the complete withdrawal of 450,000 employees from the labour market. World Economic Forum’s *Future of Jobs Report 2025* indicates that, by 2030, 32% of employees would be upskilled in their current roles, while 19% will require both upskilling and redeployment. The 11% of workers, who are unlikely to upgrade their skills, are at the highest risk. Dissatisfaction and disillusionment with state leadership (declining trust in governments and parliaments to as low as 20%) have contributed to social and political fragmentation (the Group Grievance Index has reached 7.6 on a scale of 0 to 10), polarisation and radicalisation. These are serious challenges to EU’s HC capacity. Even in the short term, HC may prove insufficient to support adequate economic growth and to maintain the required levels of security, competitiveness, and prosperity across the EU-27.

Latvia is projected to experience one of the steepest declines, decrease by as much as 21% by 2050 (Fig. 1a). Productivity is closely linked to innovation performance; Latvia is at persistently low position in EU-27 innovation rankings (Fig. 1b), which indicates low productivity.

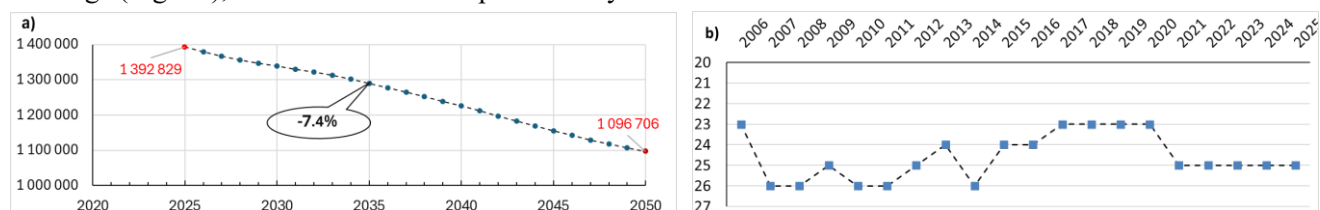


Fig. 1. Indicators characterizing Latvia’s HC; a) population aged 15-74 projections, b) innovation capability rating among EU27 countries. Sources: Eurostat, European Innovation Scoreboard.

Therefore, addressing human capital challenges at the national level, including the employability of each individual and the effective use of their full potential, is particularly relevant for Latvia. The 2025 survey *Public Engagement in Identifying Relevant Research Topics* showed that society regards human capital and labour market issues as a topical problem area in Latvia and expects scientific research to contribute to addressing them. In the survey, 45.98% of respondents assessed economic development and employment, which directly affect social welfare, as a very relevant problem area, while 29.5% assessed them as relevant. Respondents identified labour

market development and employment as important thematic priorities for research. They emphasised the need for an in-depth study of labour market structure, with particular attention to HC development as a prerequisite for sustainable economic growth. Overall, these findings indicate strong public consensus and a high level of expectation regarding practical solutions to be delivered as this Project outcomes.

The inadequacy of the number of employees and their capacity points to a challenge for HC management – to ensure that every individual: (1) is fully and sustainably integrated into the labour market, (2) possesses skills that meet rapidly changing labour market demands, and (3) has both the opportunity and motivation to apply those skills effectively. To this end, HC management should be urgently upgraded both functionally and technologically, bringing individual-level (hereinafter – micro-level) functions, services, and tools to the forefront.

However, traditional scientific and analytical methods based on the processing of aggregated datasets are of limited use for real-world micro-level HC management. To date, no established national human capital management methodologies or tools based on micro-level data are known to exist. Their development requires prior scientific-analytical studies to uncover causalities in processes in the context of data processing and determine the methodological solutions for newly developed services and tools. Unfortunately, this issue has not received sufficient attention in the research. Published micro-level studies focus on: (1) specific aspects of human capital management, such as employee capability, collaboration, employability, and labour market transitions, or (2) management within a single organisation or sector. Their purpose has not been to develop national human capital micro-level management algorithms and methodologies for the design of practical models and tools.

The objective of this research is to examine national human capital management processes and identify causal relationships in the context of advanced micro-level data processing capabilities. Specifically, the research aims to: (1) develop mutually coordinated methodologies; (2) create user-friendly prototypes of innovative tools that together form a modular human capital management system for sufficiently accurate, objective, evidence- and data-based management of HC in both static and dynamic terms; and (3) provide recommendations for public administration, thereby enabling practical impact on the country's economic growth and social development. The study will be conducted at the modelling and application levels of management science, combining quantitative and qualitative methods to generate near-optimal solutions.

Achieving this innovative objective will mean a significant scientific novelty in economic science, particularly in the national human capital management, as well as new technological know-how and innovation in data processing. The above considerations clearly demonstrate the topicality of the research not only to HC, but also to sustainable economic growth and the country's broader social development.

Given the strategic importance of national human capital development, upgrading HC management in Latvia is particularly actual. To support this process, a unique micro-level data warehouse has been created by integrating data from 30+ statistical databases into a single ontology for identifying and applying micro-level causalities and enabling the sequential development of methodologies and tools. In addition, an evidence-based short-term forecasting approach for labour market needs has been developed, drawing on employers declared demand, including vacancy data, information on actual investment and business expansion projects, and national planning documents. These technological developments provide a solid foundation for the implementation of the Project. Although their execution did not require extensive prior scientific research, several participants of the Project's scientific team were involved in their development.

To achieve the stated objective, a working hypothesis is proposed: the adaptation of advanced data processing methods, including data mining algorithms and large language models, will make it possible not only to manage national human capital reliably and dynamically at the micro-level, but also to enhance HC management with previously unattainable innovative services and tools. They will support public administration, employers, and individuals by enabling the analysis, measurement, management, and prediction of labour market processes at the level of a specific economically active resident. The contribution of informatics to management science is not a theoretical factor of this highly innovative Project; the interdisciplinary integration is a direct source of the Project's generated benefits. The adaptation of advanced data processing methods for HC management therefore represents a valuable novelty in economic science, particularly in management science.

To test and implement the hypothesis, the HC management models and tools to be developed will be based on a prescriptive human capital approach, sequentially applying the capabilities of business analytics stages to the processing of labour market datasets (Fig. 2).

For descriptive analytics, historical micro-level HC data will be used, drawn from the ontology developed within the HC data warehouse. To support effective descriptive analysis, methodologies and tools will be developed to measure the achieved aggregate capacity of national human capital and to conduct a comparative

assessment of the contribution of each HC-forming policy. This will provide comprehensive information, based on individual-level data, on the current quantity, quality, and capacity of the HC stock, as well as on the outcomes, strengths, and weaknesses of the HC-forming policies' implementation.

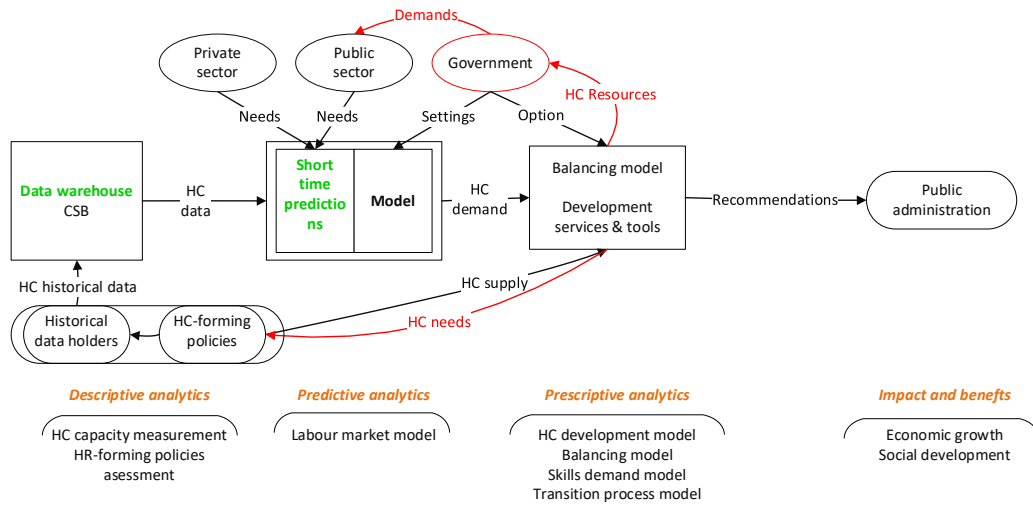


Fig. 2. Research approach: prescriptive human capital approach.

To perform predictive analytics, the developed fact-based short-term prediction model will be applied. In addition, a labour market model will be developed to translate the objectives and settings of national development planning documents into labour market data for generation of predictions of overall HC demand.

At the prescriptive analytics stage, projected HC supply data expressed in internationally recognised indicators, will be transposed into labour market data (HC development model) and compared with projected HC demand. Innovative models and tools, including a skills demand model, will be mathematically developed. The demand–supply balance will be sought (balancing model), alternative HC supply scenarios will be developed to recommend the optimal option to the government for proactive, data-driven decision-making. Subsequently, data on HC needs will be used to adjust the activities of HC-forming policies. The government will thereby gain the capacity to adjust human capital demand in the public sector on an evidence-based basis. The preferred option approved by the government will be driven with the aim of achieving the maximum possible impact on the country's economic growth and related social development under the specific circumstances.

The strategy outlined above defines the main tasks required to implement the hypothesis, ensuring that the methodologies and tools developed will provide:

- metrics for assessing national human capital capacity and its development dynamics, integrating the HC-related outcomes of all HC-forming policies;
- comparative assessment of the human capital related outcomes achieved by each HC-forming policy;
- matching HC demand with the quality and quantity of human capital available in the labour market, provision of relevant recommendations;
- matching the implementation outcomes of HC-forming policies with the human capital supplied to the labour market, provision of relevant recommendations;
- HC demand-supply balancing, provision of tasks for HC-forming policies to implement preferred option;
- assessment of labour market demand for specific skills;
- analysis and evaluation of transition processes of economically active population within the labour market.

This list of services should not be regarded as exhaustive or final; the modular design principle of the national human capital management system will enable the gradual expansion of the range of services and tools. The prototypes developed will clearly demonstrate the implementation of the Project, as the study of relevant processes and the subsequent development of methodologies will directly result in the creation of practical tools.

To achieve the set objective and complete the related tasks, coordinated research and analytical studies will be conducted, and the needed methodological basis for creation of tools' prototypes will be developed. All prototypes will be validated in a laboratory at TRL 4, ensuring testing and validation under relevant conditions. No specialized digital skills will be required to use the tools. The study will be implemented as an iterative process; an Agile business process improvement methodology will be applied throughout the research. The scientific team will need

relationship between HC supply – shaped by education, training, healthcare, demographic trends, employment, and social-cohesion policy interventions – and the demand side, driven by employers’ needs, sectoral growth, and broader economic trajectories defined in national development planning documents. By synthesising evidence from existing data sources and policy frameworks, WP1 establishes the innovative conceptual and empirical basis for the subsequent modelling work packages. Its outputs feed directly into the Project’s development phase, providing the know-how and analytical foundation required for WP6 (the labour market model), and WP7 (the human capital supply model), while also contributing coordination inputs to the wider work-package network. *WP1* is expected to identify key imbalances, bottlenecks, and leverage points within the HC management system, generating actionable insights that will inform both the design of quantitative models and the formulation of policy recommendations in later stages of the Project.

WP2 establishes an innovative methodological framework for measuring HC capacity and assessing the contribution of each HC-forming policy to the capacity achieved. Current methods tend to focus primarily on education and healthcare, without adequately capturing the quantitative dimensions of human capital and the effects of societal fragmentation. In addition, the selection and weighting of indicators are still largely defined subjectively. The WP2 will address the limitations of existing approaches to measuring HC capacity. A key task of WP2 is therefore to develop an innovative methodology for the objective mathematical computation of key performance indicators (KPIs) of each HC-forming policy and modelling their contribution to the overall HC capacity. Coordination between WP1 and WP2 will ensure common alignment with HC-forming policies (sectors), regarding the indicators to be used.

WP3 comprises research on the actual demand for cognitive skills in the labour market. Existing studies on the growing demand for cognitive skills consistently emphasise their primary importance but often pay insufficient attention to professional skills and sector-specific differences. The HC data warehouse has accumulated a substantial volume of information, providing a unique basis for an evidence-based study of the professional skills required by employers and the dynamics of this demand. The research includes the development of an innovative, empirically validated classification framework for cognitive skills, as well as the analytical identification of common combinations of cognitive skills. This will reveal typical skill bundles and their relative importance and demand intensity. The results will provide a basis for identifying mismatches between the skills demanded by employers and those supplied by the workforce under WP9.

WP4 covers research and analytical work on individuals’ transitions in the labour market changing their status depending on profession, industry, age, education, gender. WP4 also provides a basis for forecasting major labour market changes resulting from the green and digital twin transitions, the introduction of artificial intelligence and robotics, as well as the declining number of employees. The HC data warehouse offers a unique opportunity to analyse accurate and unbiased data, enabling data-driven identification of groups of employees, who are actively engaged in upskilling and reskilling and preparing for labour market changes, as well as those who remain passive. To reveal how projected labour market changes will affect specific professions and sectors, how faster transition flows could reduce unemployment, how labour market transitions affect individuals’ quality of life – these are only some of functional issues that can be prepared by WP4 for WP10 (HC transition model).

Work packages WP5-WP11 cover the development of models and tools prototypes at the application level, based on the methodologies created in the research work packages WP1-WP4.

WP5 develops an innovative labour market model. Its objective is to model the required quantitative and structural dynamics of the labour market by linking the economic settings in national development planning documents, such as GDP growth, with the labour market outcomes needed to achieve them – the number of employed persons in the country as a whole and in territorial units, by profession, health status, education level, and skills profile. The model will support the assessment of different development scenarios, for example by increasing labour productivity and/or increasing the number of employees under various assumptions. The model receives the necessary know-how from WP1 and labour market data from the HC data warehouse. After modelling, the resulting data on required HC will feed into WP8, where HC demand is balanced with potential HC supply. In accordance with the Agile methodology, model development will be followed by a dedicated phase of testing and refinement to ensure robustness, credibility, and practical applicability.

WP6 develops the HC development model, a supply-side computational model that simulates how the HC stock is formed over time. The HC supply is characterized by a broad set of internationally agreed outcome indicators of HC-forming policies. The model is designed to project how current policies’ outputs shape HC supply, providing an innovative, dynamic, and forward-looking picture of HC in the labour market under varying policy scenarios. WP6 situates the supply modelling within the conceptual and empirical findings established in

the Project's foundational research work package WP1. As with WP5, the WP6 includes a dedicated phase of testing and iterative refinement to validate the model's behaviour and ensure its outputs are fit for cross-model integration.

WP7 operationalises the HC capacity measurement methodology developed in WP2. The relationships between variables will be carefully modelled to determine whether, and under what conditions, the observed relationships can be interpreted as causal. Using computational algorithms, statistically significant KPIs will be calculated, and a mathematical expression of the HC capacity index will be developed to measure achieved HC capacity in both statics and dynamics, and to observe progress and deficiencies in separate aspects. Model testing will assess the stability and sustainability of the model. In accordance with the Agile methodology, the modelling procedure may be repeated, including using alternative computational tools to improve the results.

WP8 develops the central innovative integrative quantitative model of the Project, designed to balance HC supply and demand across the labour market. Drawing on supply-side and demand-side inputs received from WP5 and WP6, the model directly engages in cross-sectoral balancing; it reconciles diverging trajectories between the skills and competencies that the labour market requires, and the HC amount that policies are capable to produce. The balancing model operates as the analytical core of the entire modelling architecture, aggregating outputs from upstream work packages into a coherent, cross-sectoral quantitative framework. The primary task of WP8 is to translate this quantitative reconciliation into concrete, evidence-based recommendations to HC-forming policies and development planning documents. By simulating the consequences of different policy scenarios – covering education, retraining, migration, healthcare, cohesion, employment – the model identifies which combinations of supply-side and demand-side measures are most likely to close observed gaps in HC. Its outputs, including recommendations, tasks, and options for correcting plans and policies, provide policymakers with a rigorous, model-grounded basis for innovative strategic decision-making in HC development. At the same time, the supplied balanced data also contribute to the development of project outcomes in WP9 and WP10.

WP9 develops a comprehensive model of professional and cognitive skills demand, building on the analysis of employers' requirements conducted in WP3 and on further applied research. Categories of skills will be aligned with ESCO. *WP9* will define and operationalise the core cognitive skills that are critical for the future labour market, including analytical and critical thinking, problem-solving, active learning, creativity, leadership, and others. The WP will include the selection and adaptation of suitable measurement instruments, as well as the development of a methodology for their testing and validation. Following thorough testing, the methodology will be applied and iteratively refined with participants from representative fields, including individuals with different levels of professional experience and educational backgrounds. The resulting data will be used to further calibrate the model, establish dynamic and profession-specific benchmarks, and identify gaps between current skills supply and labour market demand. This will generate actionable and forward-looking insights for policy and practice.

WP10 creates a graph model of transitions in the labour market according to functional issues developed in WP4. HC data warehouse contains data on current and past situation, while labour market projections and balanced data from WP8 provide forward-looking labour market data enabling prediction of transitions. Using analytics performed in WP4, the model is intended as a tool for proactive coordinated labour market policy management, including public employment and career services, targeted training system and talent availability, fact- and data-based decision making.

WP11 develops a tool for evaluating the capacity of HC-forming policies and their contribution to HC strengthening. Using the methodological framework established in WP2 and data on the implementation outcomes of HC-forming policies, WP11 will produce a data-based numerical assessment of each policy's contribution to the quantity and quality of the achieved HC capacity. The identification of weaknesses in specific policy areas will also enable the identification of priority lines of action. Comparative assessment of policy contributions will make it possible to formulate targeted recommendations for improving performance of less effective policies.

The achievement of the Project objective will be demonstrated by: (1) models and tools created within the Project, and (2) quality parameters of these models, including, but not limited to, coefficient of determination R^2 , p-value, standard deviation, RMSE, MAE, confidence intervals, F-statistics. The applicability of model and tool prototypes, and consequently the validity of the methodologies developed, will be tested in a laboratory environment using a specially designed set of virtual data compositions.

The knowledge, skills, and capacity of the scientific team are fully adequate for the high-quality execution of the research tasks and the achievement of the goals within the planned timeframe. The Project represents a further development of studies conducted by participants of the scientific team in previous years; they have long-standing experience in fields closely related to the Project thematics, as demonstrated by their publications and h-indexes.

The results of previous research are confirmed by 40+ publications on the study issues, 25 of which are indexed in Scopus and/or WoSCC. The team has accumulated practical experience through parallel work in and/or for public administration, as documented in the attached CVs. This combination of scientific expertise and practical policy experience provides a strong basis for the successful implementation of the Project.

To date, no comparable HC management tools based on micro-level data, nor equivalent adaptations of advanced data processing methods for HC management, are known to exist. The innovative approach developed within the Project will therefore provide a significant practical contribution to optimising HC management in support of intensified national development: (1) more effective HC management by state institutions and municipalities, (2) innovative services for businesses and residents, (3) measurement of HC capacity and evaluation of the contribution of HC-forming policies, (4) balancing of national development objectives with the capacity of HC-forming policies, and (5) access for institutions, businesses, and residents to objective, sufficiently accurate, and easy-to-use fact- and data-based information on HC in both static and dynamic terms.

2. Impact

2.1. Scientific results and technological findings of the project, the plan for their distribution

The results obtained will confirm the achieved serious scientific novelty in economic science, particularly in the field of national human capital management, as well as new technological know-how and innovation in data processing methods. In addition to its research contribution, the Project will deliver practical outputs in the form of TRL4 level prototypes of models and tools, as well as recommendations to the public administration.

The Project will develop mathematically defined and/or computer-based algorithms and methodologies for micro-level HC management by adapting advanced data-processing methods, including data-mining algorithms and large language models. The transition from traditional macro-level HC data processing methods to micro-level algorithms will generate benefits, which would substantially impact on the country's socio-economic development in an increasingly complex and volatile global environment. In particular, it will create: (1) opportunities to manage HC reliably and dynamically, and (2) the possibility to enhance HC management with innovative services and tools that were previously unattainable:

- enabling public administration to: (1) manage HC at the level of a specific economically active resident, (2) analyse, measure, manage, predict labour market, (3) evaluate overall HC capacity and the contribution of each HC-forming policy, and (4) develop national and sectoral planning documents based on HC data;
- considering employers' requirements into national human capital management;
- providing every individual with opportunities to: (1) assess their sustainable employability prospects; (2) make maximum use of their potential; and (3) contribute effectively to the country's economic growth and social development.

The technological findings of the Project will be a direct result of the research conducted, with the outputs of modelling-level WPs feeding directly into the application-level WPs. Two types of practical results are expected:

1. Prototypes of appropriate, user-friendly, innovative tools will be developed at TRL4 to support sufficiently accurate and objective experimental, fact- and data-based HC management in both static and dynamic terms:
 - a) measurement of HC capacity and its development dynamics, integrating the HC-related outcomes of all HC-forming policies;
 - b) comparative assessment of the contribution of each HC-forming policy in HC capacity;
 - c) matching HC demand with the quality and quantity of HC available in the labour market;
 - d) matching the implementation outcomes of HC-forming policies with the HC supplied to the labour market;
 - e) balancing HC supply and demand within the labour market;
 - f) assessment of labour market demand for specific skills;
 - g) analysis and evaluation of HC transition processes within the labour market.
2. Recommendations developed for public administration institutions:
 - a) helping to align policy objectives with actual HC capacity when developing national development planning documents, thereby increasing the likelihood of their successful implementation and strengthening public trust in public administration;
 - b) identifying weaknesses to be minimized in the relevant area when designing and implementing HC-forming policies, thereby increasing overall HC capacity and enabling the achievement of more ambitious yet realistic national socioeconomic development goals.

Both lists should not be considered exhaustive or final; the modular design principle of the HC management system will enable their gradual expansion.

The planned dissemination of the Project's scientific results and technological findings is aimed at generating impact within the broader scientific community, fostering scientific cooperation, and ensuring the applicability and sustainability of the knowledge produced.

The theoretical deliverables of the study will be published in peer-reviewed scientific journals indexed in Scopus and/or WoSCC following Open Access principles. It is planned to submit five original research articles presenting the Project's results in the following areas: (1) development of a national human capital management system based on micro-level data, (2) matching HC demand and supply in the labour market, (3) HC metrics and the contribution of HC-forming policies, (4) development of a validated cognitive skills demand framework, and (5) development and validation of a skills demand model. The articles will present the relevant datasets, mathematical expressions of the models, descriptions of the modelling methodologies, and the tools developed. Appropriate thematic peer-reviewed Open Access journals will be selected, such as *Human Resource Management Review*, *Economic Analysis and Policy*, *Economic Research-Ekonomska Istraživanja*, *Intellectual Economics*, *Journal for Labour Market Research*, *Data Mining and Knowledge Discovery*, *Mathematics*, *Cognitive Systems Research*, *Journal of Economic Psychology*, *Cognitive Processing*. Before publication in journals, all article preprints will be made publicly available in the University of Latvia (UL) E-resource repository (<https://www.biblioteka.lu.lv/en/search-and-find/e-resource-repository/>).

Two intellectual property rights on HC capacity assessment issues and modelling of HC supply-demand balancing will be prepared for registration with the International Certification Authority *Copyright.eu* (<https://www.copyright.eu/en/>). To ensure access to the research data generated, the scientific databases and datasets prepared in accordance with the FAIR principles, including metadata, will be deposited in the internationally certified research data repository *DataverseLV* (<https://dataverse.lv/en/>).

The conceptual approach and research results of the Project will be presented at three international scientific conferences on management science and/or human capital issues organized in the Baltic States or in the nearby region, as they are expected to potentially attract greater participation of Baltic scientists working on HC related issues. Existing analyses, such as the *PwC Baltic Human Capital and Work Environment Survey 2025*, indicate that similar HC related challenges are present across all three Baltic countries.

Following the *OECD Recommendation on International Cooperation in Science and Technology*, the Project will initially be presented to strengthen networking with Baltic university researchers. This will help identify shared challenges and priorities that may lead to joint, evidence- and data-based international research on various aspects of micro-level HC management, including employability, the impact of ageing on the labour market, returns on investment in HC formation, commuter migration, regional dimensions. Such cooperation may also support the preparation of a new international project proposal for a Horizon Europe call.

Presentations at the three annual international scientific conferences of the UL will demonstrate progress in the Project's development and highlight innovation in HC management, with the aim of encouraging researchers from various scientific fields to engage in similar research activities. Regular internal seminars at the UL will also be organised as creative discussions, involving specialists from different areas of economics and informatics.

Overall, the wide distribution of generated new knowledge will provide, among other benefits: (1) broad dissemination and accessibility of the study results, (2) opportunities to adapt the Project concept for other purposes, (3) the initiation of follow-up studies, and (4) the establishment of new contacts and the expansion of scientific cooperation. The planned calendar of dissemination events is included in the Activity Plan (Section 3.2).

2.2. Socio-economic impact and publicity of results

The Project outcomes will include TRL4 level prototypes of tools that have been validated in a laboratory environment and are ready for experimental use, which will also support validation in a relevant environment, contributing to their further development for operation under realistic conditions. Several directly applicable Project outcomes, shown in red in Figure 2, could have a practical impact on the entire society if used by:

- the Ministry of Economics, which is responsible for overall national human capital management;
- national and local public administration institutions, when:
 - ✓ developing national and local socioeconomic development planning documents to define near-optimal policy objectives in line with available HC capacity;
 - ✓ designing HC-forming policies and assessing their contributions to overall HC capacity;
 - ✓ identifying and assessing HC transitions;
 - ✓ comparing HC capacity and the contribution of HC-forming policies with those of other EU-27 countries to adopt best practices;

- private and public sector employers, when defining their planned workforce needs, which will feed into the HC management system and by influencing skills-demand and transition indicators will enable HC-forming policies and individuals to respond to these needs in a timely manner;
- residents of the country, when identifying demand for their skills and making timely decisions on the need for, and direction of upskilling or reskilling.

The Project results meet the expectations expressed by society. HC and labour market management at the individual level will provide every person with the opportunity to seek employment in line with their skills and capacity. Individuals will be able to regularly assess the alignment of their skills with labour market requirements in the coming years and determine the necessity and urgency of upskilling/reskilling, thus increasing their contribution to the country's GDP growth and, consequently, improving their quality of life.

Intellectual property rights will be registered and maintained in accordance with the *Procedure for Identification, Evaluation, Registration and Maintenance of Intellectual Property at the UL*.

Regular public information procedures will be carried out aimed at a specific target audience to popularize contributions of the Project to the society (see Activities Plan): (1) seminars/discussions for responsible public administration institutions and business organizations with the aim of informing about new services in the field of HR management and their use, (2) quarterly updated project websites on UL/FST and IMCS portals, and (3) popular informative articles on social media about the opportunity for every individual to identify his/her skills demand and to make timely decisions about the need and direction of upskilling/reskilling.

2.3. Contribution to the capacity building of the scientific team of the project, including students, and to the improvement of the study environment

The Project provides a strong platform for developing the scientific capacity of both established researchers and the next generation of scientists. Its implementation will strengthen scientific competence of team's participants in adapting advanced data processing methods to improve the applied performance of economics science, particularly HC management. This will be achieved through the development of innovative research solutions, including methodologies, models, and tools for services that were not attainable prior to the Project. The Project outcomes also have the potential to inspire scholars to modernise traditional management methods in other subdisciplines of economics and even in other socioeconomic processes.

Students involved in the Project will gain experience in applied research, particularly in adapting and applying advanced data processing methods and technologies to specific tasks. They will also improve their analytical competences and develop skills in preparing scientific publications and disseminating research results. The principal investigator will consult the MS student on the data processing issues, while the lead participant J. Skilters will supervise the doctoral student in the field of cognitive issues.

The methodologies, datasets, and tools' prototypes developed within the Project will contribute to improving the study environment at the FST by enriching course curricula with new practical case-study materials, e.g. in the courses DatZB068 "Databases and Information Systems Fundamentals", DatZM037 "Applications of Language Technology", and in the newly developed course "Data Mining".

3. Implementation

3.1 Applicant and Scientific Team

A key advantage of the UL is its capacity to conduct interdisciplinary research by involving experts from various fields in scientific teams. The essence of this Project is to elevate management field of economics science, particularly national human capital management, to a qualitatively higher level to promote national socioeconomic development. This will be achieved by adapting and integrating modern data processing methods, which belong to the research field of computer science and informatics. Therefore, the core of the scientific team consists of experienced researchers from the Faculty of Science and Technology (FST) of the UL, whose competences include the application of advanced data processing methods and tools for various socioeconomic tasks. To implement the Project, it will be possible to involve researchers from other scientific fields, including management science, statistical analysis, econometrics.

At the same time, when creating the Project research design, the purposefulness was identified to supplement the competence of applying artificial intelligence for performing certain activities, that would be difficult for FST researchers alone. The skills and experience acquired by researchers from the UL Institute of Mathematics and Computer Science (IMCS) in developing large language model applications and a short-term prediction model of labour market will be valuable for the development of the labour market model, the HC development model, and the balancing model.

International Evaluation of Latvia's Scientific Institutions Activity, conducted by Technopolis in 2025, assessed the UL as a “strong international player showing improving research performance and possibility to advance incrementally”. The FST was rated “very good” for research quality, achieving outstanding performance results. The evaluation highlighted notable technological developments and active involvement in national policy development, including strong links with decision makers. During the assessment period 2019–2024, FST delivered 87 computer programs and algorithms and participated in the development of national and practice-oriented information systems. FST’s rating in the economic impact category is “good”; the Project is expected to make a significant contribution to further improving this rating. The faculty’s modern research environment and infrastructure were rated as very good; it is suitable for processing large volumes of data.

The Technopolis Expert Group recognised IMCS as a strong national research performer, with a good level of research and high-quality outputs in several areas of clear international visibility. These areas include artificial intelligence, real-time systems, language models, and text recognition; the latter being strongly supported by research in computational linguistics. The Institute’s contribution to cybersecurity was also noted. IMCS has substantial experience in cooperation with public administration institutions, including the CSB, which is essential for the implementation of this Project. Recently, IMCS has acquired up-to-date tools for real-time and autonomous systems research, which are necessary for the execution of the Project.

Project participants have necessary technological equipment at their disposal. A critical element of the Project's research infrastructure is the high-performance computing environment of the CSB, where the HC data warehouse is maintained. CSB operates a dedicated server equipped with 1TB RAM and L40S GPUs, providing the computational capacity required to run state-of-the-art large language models (such as Gemma 4) entirely on-site. This configuration ensures that all processing of HC micro-data takes place within CSB's secure, air-gapped environment, in full compliance with the *Statistics Law* and the *General Data Protection Regulation*, eliminating any need to transmit sensitive data to external cloud. This infrastructure makes it technically feasible, for the first time, to apply advanced LLM-based analytics directly to individual-level labour market data at national scale.

The Project builds on the previous successful scientific and analytical activities of the lead Project participants (see CVs in Chapter 2 of Part A). They have collaborated for many years and across numerous projects, including both large interdisciplinary scientific- and practice-oriented initiatives. The assignment of tasks and the division of responsibilities among the lead Project participants, based on their qualifications, as well as the partial overlap of competences required for effective collaboration, are clearly presented in the Activity Plan (Section 3.2). The competences of the lead Project participants cover the key aspects of the research: (1) G. Karnitis – data-mining methods, ontology development, modelling of socio-economic processes, public-administration functionality, (2) J. Skilters – research on cognitive and intelligent systems, (3) G. Barzdins – data warehouse ontologies, artificial intelligence, large language models, and (4) E. Karnitis – human capital issues, data and system analytics and integration, strategic planning. Overall, this provides strong confidence that: (1) the stated objective will be achieved even in the event of potential risks (see Risk Plan Chapter 3.3), (2) the scientific results will make a meaningful contribution, and (3) the practical outputs will be attractive and usable for the intended users

Project management is entrusted to an experienced principal investigator who has successfully participated in several studies and projects, including leading scientific teams in the successful implementation of two projects of LCS Fundamental and Applied Research Programme in the 2020 and 2021 project calls. Professor G. Karnitis has conducted research in fields directly related to the Project, which strongly confirms his personal expertise, know-how, and capacity to lead the Project effectively.

A shared working environment will be implemented using the UL’s subscribed MS OneDrive service, which is accessible exclusively through authenticated identification and authorisation. Within this platform, a shared directory is established for storing Project related working documents. Access to this directory will be granted and maintained only for active Project participants. This approach will ensure the secure management of both received information and information generated within the Project in a controlled collaborative environment.

The research data obtained, linked to the Project results and outcomes and prepared in accordance with the FAIR principles, including metadata, will be deposited in the internationally certified research data repository *DataverseLV* to ensure general accessibility. A research data management plan will be developed and maintained with the involvement of a member of the UL data steward team to guarantee the integrity and proper structuring of both the datasets and their associated metadata. The ARGOS platform (<https://argos.openaire.eu/portal/>) will be used following the recommendations of the LCS. The plan will be submitted to the LCS within three months of the start of the Project (see Activity Plan) and deposited in the Zenodo research data repository (<https://zenodo.org/>).

The use of funding is planned with priority given to mandatory information and publicity costs, considering potential inflation. The remaining funding is allocated to remuneration and coverage of indirect eligible costs (see Chapter 4 of Part A).

The Ministry of Economy is interested in the successful implementation of the Project for the practical application of its results. If the Project application will be approved, ME will support the implementation of the planned Project tasks by providing Project participants with access to a secure data processing environment and to the HC data warehouse established at the CSB, an institution within the ME system. ME will also allocate additional funding for the implementation of the Project.

3.2 Activity plan

The current Project will be implemented over a period of 36 months (Tab. 2), from 1 January 2027 to 31 December 2029. The project execution is divided into sequential work stages, following *The Standard Project Management Life Cycle*. Tasks performed by the Project partner IMCS, connection of Project results and outcomes (numbering from Section 2.1) with WPs generating them as well as schedule of publishing and publicity events also are shown in the Plan. Manager of each activity is identified.

Activity Plan

Table 2

[illegible]

Outcomes according to section 2.1.  Preparation, monitoring, updating information and data, iterations  Implementation of the activity

A - article; I - intellectual property application; U - UL conference; C - external conference, trip; P - WWW page; S - seminars, discussions, etc.

The Initiation Stage is implemented by preparing the Project's application, it is not shown in the Plan. The Planning Stage includes detailed aspects of project implementation planning. The Execution Stage involves the sequential and parallel execution of 11 interconnected WPs (Section 1) to complete the project tasks and achieve the project objectives. The methodology for carrying out each WP is described in Section 1. The execution of activities will partially overlap to reduce potential risks. The application of the Agile business-process improvement methodology will further reduce potential risks related to the generation of new ideas and the implementation of changes through iterative cycles during the Stage. The Control Stage will include comprehensive testing of the prototypes and iterative refinements to verify their quality and ensure achievement of TRL4 level. Project closing documents are being prepared at the Closure Stage.

No purchasing new fixed assets, tools or materials, nor use of external services are planned for the implementation of the Project. The average available remuneration of a scientific team participant for Project implementation is 666 EUR per month, including employer and employee taxes, which is extremely low for high-skilled researchers (see Chapter 4 of Part A).

3.3 Project management and risk plan

Project management will be carried out in accordance with the guidelines of the *Project, Programme and Portfolio Management standard ISO 21500:2021*, as adapted in Latvia, and the *Regulations on the Implementation of Projects at the UL*. The principal investigator has successfully led scientific teams in the implementation of LCS projects lzp-2020/2-0073 and lzp-2021/1-0108. Each lead Project participant has managed several projects, which provides relevant experience and a clear understanding of the principal investigator's functions and responsibilities in ensuring successful Project implementation.

Project management will be implemented in an analogous, practically proven manner – weekly online meetings of the entire scientific team (including Partner) to discuss the work completed and specifying tasks. Through teamwork among experienced colleagues, potential problems can be addressed effectively. This has already been demonstrated by the successful completion of the Project's initiation phase within the limited timeframe available.

Intellectual property management will be conducted in accordance with the *Procedure for the Identification, Evaluation, Registration and Maintenance of Intellectual Property at the UL*.

In the opinion of the Project applicant, there are no high-probability risks that could jeopardize the implementation of the Project (Tab. 3). The approach for minimizing the impact of potential risks is presented therein. All risks will be reduced primarily through careful research planning, as well as through partial overlaps in both the scheduling of work packages and the competences of Project participants. The application of the Agile methodology will further reduce potential risks associated with the generation of new ideas and the introduction of changes through iterative cycles during Project implementation.

Risk Plan

Table 3

	<i>Risks</i>	<i>Description of the risk</i>	<i>Assessment</i>		<i>Causes and/or prevention / mitigation measures</i>
			<i>Likely-hood</i>	<i>Impact</i>	
1.	Scientific risk	Insufficient qualification of the scientific team for development of methodologies and algorithms	1	3	Scientific team expansion with highly qualified researchers
		Models and tools quality is insufficient for their practical use	1	2	Activated iteration according to the Agile approach
2.	Implementation risks	Outflow of highly skilled participants	1	3	Overlaps in critical competencies are actively exploited, attracting other researchers are sought
		Insufficient motivation of students	2	1	The environment for acquiring knowledge and skills is improved, motivating tasks are assigned
3.	Result achievement risks	Significant delays in WP execution affecting subsequent WPs	2	3	Replanning and rescheduling of Activity Plan, more intensive use of parallel activities
		Failing to achieve planned results	1	2	Evaluation of causes, replanning of activities and resources
4.	Management risks	Failure of the scientific team to cooperate	1	3	Workshops, discussions, etc., to increase transparency of information sharing and collective responsibility for results
5.	Legal risks	Violation of project conditions	1	1	Analysis of the causes of violations, their prevention measures
6.	Financial risks	Unforeseen need for outsourcing to increase expertise in a specific area	1	2	Work plan and funding allocation are replanned and updated, additional funding is being sought

When developing the Project description, ChatGPT 5.5 tool was used to improve grammar and lexicon.